AGV2 Kayaba — Function Summary

Differential-drive magnetic-tape AGV with front and rear MGS1600 sensors, RFID route tags, trolley-hitch pusher, lidar safety fields, keyboard / remote control and a Flask web HMI. Read from `profiles/agv2_kim.json` and the controller code (commit 85d18b0) · 15 Oct 2026. All RFID tag values are placeholders until replaced with the real scanned hex.

1. Operating modes

Mode	Entered by	Behaviour
ARMED	Startup, RESET, arriving at homepost or B0	Stopped, motors released, waiting for a start command.
RUNNING (AUTO forward)	Branch button DI4–7 / keys 7 8 9 0, CALL START (DI8), web START, or end of a reverse	Forward PID tape-following on the front MGS1600.
REVERSING	Tag 0001 (branch 2: tag 0002), tag 0003 (not branch 2, left track), web START REVERSE, or key R	Backward PID tape-following on the rear MGS1600.
MANUAL	Mode selector DI2 ON, key Q, or web bypass	Driving from the web jog pad, the laptop keyboard or the USB remote (WASD / arrows).
ROTATING	Key T / Y or web ROTATE 180° (from MANUAL or ARMED)	180° spin-turn on the tape, then braked back in MANUAL / ARMED (section 11).
EMERGENCY	E-stop DI3, key E, or web E-STOP	All motion stops, motors released (AGV can be pushed). RESET (DI1 or ENTER) to recover.

2. Ways to start and stop

- **Branch buttons DI4–7 or keys 7 / 8 / 9 / 0** — in ARMED: select TITIK 1–4 **and** start AUTO (normal run). In other modes they only select; ignored while driving.
- **CALL START (DI8)** — only at homepost (DO11 ON). That run stops at tag B0; continue with a branch button or web START.
- **No physical START button** — DI0 is the pusher button (`DI_START: -1`).
- **RESET** — DI1, the ENTER key / OK on the remote, or web RESET (bypass mode). Leaves EMERGENCY; aborts AUTO / REVERSE back to ARMED.
- **Web MANUAL page** — bypass mode gives START / RESET / MODE / E-STOP buttons (no password); START REVERSE starts REVERSE from ARMED.

3. Speeds

Mode / setting	Value	Notes
APPROACH (tag 0012, key 1)	0.1 m/s	APPROACH PID gains, eases down at 0.2 m/s²
SLOW (tag 0014, key 2)	0.2 m/s	SLOW PID gains
EXTRA_SLOW (after a reverse)	0.15 m/s	SLOW PID gains · set by the hitch stop (00CC / marker) when driving forward again; `AUTO_TARGET_EXTRA_SLOW_SPEED` on PARAMS
SLOW in branch (tags 00AA / 00AB / 00AC)	0.2 m/s	**Own SLOW BRANCH PID gains** (`KP/TD/N/V_RED_COEF_SLOW_BRANCH`, start = SLOW values, own auto-tune option) until the next speed tag or resume
MEDIUM (tag 0015, key 3)	0.4 m/s	**Own MEDIUM PID gains** (start = SLOW values: KP 2.2, TD 0.14, N 12, V-Red 2)
HIGH (tag 0016, key 4)	0.6 m/s	HIGH PID gains (KP 1.4, TD 0.12, N 20, V_RED_COEF 4)
MAX (tag 0018, key 5)	1.0 m/s	**Own MAX PID gains** (KP_MAX 1.0, TD_MAX 0.15, N_MAX 20, V_RED_COEF_MAX 4) · motors top out at ~0.94–0.98 m/s
Reverse	0.15 m/s	0.1 m/s after tag 00CA · 30 s safety timeout (paused while holding)
AUTO accel / decel	0.2 / 0.2 m/s²	ACCEL_RATE / DECEL_RATE
Manual high / slow	0.4 / 0.2 m/s	Slow speed used for turning
Manual accel / decel	0.8 / 0.8 m/s²	Smooth start and stop; direction change passes through zero
Lidar outer / middle	60 % / 30 % of the current speed	e.g. MAX 1.0 → 0.6 / 0.3 m/s · eases down at 0.2 m/s² (lidar is OFF by default)

- Keys 1–5 set the AUTO speed mode in ARMED (starting speed of the next run) or while RUNNING; the next speed tag overrides them.
- Forward steering direction is flipped (`ERROR_SIGN_FORWARD: 1`).
- **Accel / decel rates are real m/s²** (fixed 19 Oct 2026): the ramp now uses the real time between sensor frames (~10 Hz). Before, it stepped a fixed 10 ms per frame, so every ramp was ~10× slower than set (decel 0.5 took ~4.5 m from HIGH to SLOW). Braking distance = $(v_1^2 - v_2^2) / (2 \times \text{decel})$: HIGH→SLOW at 0.1 m/s² = 1.6 m, at 0.5 m/s² = 0.3 m. With a 2.1 t trolley keep DECEL_RATE ≈ 0.1.
- Tape loss or sensor (CAN) timeout: brake, resume automatically when the tape is seen again.
- **CAN auto-reconnect**: no frames from a sensor for 0.3 s → NMT start re-sent (sensor rebooted after a supply dip); no frames from any sensor for 2 s → the CANable adapter is closed and reopened (retry every 0.5 s). Logged as [CAN] no data … / data back after x s. Profile `can_sensor.NMT_RESEND_S / REOPEN_S / RETRY_S`. The 1 s safety stop is unchanged.
- Near top speed the steering difference is kept: if a wheel would need more than 5 V, both wheels are slowed by the excess (MAX 1.0 m/s asks ~3180 rpm, the motors reach ~3130).
- Speeds, ramps and PID gains can be tuned on the PARAMS page (engineer login). **APPLY saves them as the new defaults** (kept after restart); RESET TO PROFILE goes back to the profile values.

4. RFID mapping

Tag	Sequence	Fires when	Action	Cooldown
0012	speed_zone_approach	RUNNING	Speed → APPROACH (0.1 m/s)	0 s
0014	speed_zone_slow	RUNNING	Speed → SLOW (0.2 m/s)	0 s
0015	speed_zone_medium	RUNNING	Speed → MEDIUM (0.4 m/s)	0 s
0016	speed_zone_high	RUNNING	Speed → HIGH (0.6 m/s)	0 s
0018	speed_zone_max	RUNNING	Speed → MAX (1.0 m/s)	0 s
000C	release_at_c	RUNNING, **not on branch 2**	Trolley release — section 7	5 s
001C	release_at_1c_branch2	RUNNING + **branch 2 only**	Trolley release — section 7	5 s
00AA	branch_titik_1	RUNNING + branch 1	SLOW (branch gains), track left	5 s
00AB	branch_titik_2	RUNNING + branch 2	SLOW (branch gains), track right	5 s
00A2	branch2_track_left_a2	RUNNING + **branch 2 only**	Track left (speed unchanged)	5 s
00B2	branch2_ignore_ae_b2	Any mode + **branch 2 only**	Tag 00AE ignored (forward and reverse) until homepost 0000	3 s
00AC	branch_titik_3	RUNNING + branch 3	SLOW (branch gains), track left	5 s
00A3	branch3_track_right_a3	RUNNING + **branch 3 only**	Track right, like BB (speed unchanged)	5 s
00AD	branch_titik_4	RUNNING + branch 4	SLOW only (no track action yet)	5 s
00AE	fork_always_right	RUNNING, skipped after 00B2	Track left	5 s
00BB	branch_merge_reset	RUNNING (every run)	Track right (branch is **kept**)	5 s
00AE	reverse_fork_ae	**REVERSING**, skipped after 00B2	Reverse track left (rear sensor) — branch 2, 4 / no branch	5 s
00BB	reverse_merge_bb	**REVERSING**	Reverse track right (rear sensor) — branch 2, 4 / no branch	5 s
0001	reverse_at_1	RUNNING, **not branch 2**, skipped if ignore_rfid_1 set	Stop, wait 2 s, start REVERSE	3 s
0002	reverse_at_2_branch2	RUNNING + **branch 2 only**, skipped if ignore_rfid_2 set	Stop, wait 2 s, start REVERSE: starts right, **00AE → left**, 00BB → right	3 s
0003	reverse_at_3_left	RUNNING, every branch **except 2**, skipped if ignore_rfid_3 set	Stop, wait 2 s, start REVERSE on the **LEFT** track only (branch rule and AE/BB ignored)	3 s
00BA	arm_reverse_marker_ba	REVERSING	Arm DI10 stop marker (marker ignored before BA)	3 s
00CA	reverse_approach_ca	REVERSING	Reverse speed → 0.1 m/s	3 s
00CC	stop_at_c1	REVERSING (also DI10 marker after 00BA)	Hitch sequence — section 6	5 s
00CE	clear_ignore_at_cr	Any mode	Clear ignore_rfid_1 / 2 / 3 (tags 0001 / 0002 / 0003 work again)	3 s
00CB	ignore_reverse_tags_cb	Any mode	Set ignore_rfid_1 / 2 / 3 — reverse tags 0001 / 0002 / 0003 ignored until 00CE	3 s
0000	arrive_homepost	RUNNING, not right after leaving home	Stop, **clear branch**, re-arm tag 00AE, exit to ARMED, DO11 ON	5 s
00B0	stop_at_b0	RUNNING, CALL START runs only	Stop, exit to ARMED	5 s

- **Branch selection** (TITIK 1–4) stays set for the whole run and is cleared **only at homepost** (tag 0000, or reading the homepost tag while parked).

- Cooldown only stops the **same** tag firing twice in a row; it never delays the first read. While a stop sequence is running (C, 1C, CC, 1) other tags are ignored.

5. IO mapping

DIO module 192.168.1.31 (Modbus TCP, 16 DI / 16 DO). All DIs are inverted in software (`DI_FLIPPED = 1`): "ON" below is the value the code sees and the IO page shows; an unwired input reads ON.

Digital inputs

DI	Function	Meaning / use
0	PUSHER BUTTON	Press: full UP / full DOWN by position; press while moving = stop. ARMED or MANUAL only
1	RESET	Rising edge = reset / leave emergency (same as ENTER key)
2	MODE selector	ON = MANUAL, OFF = AUTO (key Q overrides until the selector is moved)
3	EMERGENCY	ON = e-stop tripped
4	BRANCH 1 button	Select TITIK 1 (+ start AUTO in ARMED)
5	BRANCH 2 button	Select TITIK 2 (+ start AUTO in ARMED)
6	BRANCH 3 button	Select TITIK 3 (+ start AUTO in ARMED)
7	BRANCH 4 button	Select TITIK 4 (+ start AUTO in ARMED)
8	CALL START	Start AUTO only at homepost; run stops at B0
9	TROLLEY LIMIT 1+2 (NC)	Both switches in parallel · OFF = trolley caught
10	REVERSE STOP MARKER	Rising edge while REVERSING = precise stop
11	LIDAR OUTER (NO)	ON = object · 60 % of current speed (when lidar enabled)
12	LIDAR MIDDLE (NO)	ON = object · 30 % of current speed (when lidar enabled)
13	LIDAR INNER (NO)	ON = object · stop at once (when lidar enabled)
14	PUSHER PROX TOP	Pin at top
15	PUSHER PROX BOTTOM	Pin at bottom · ends FULL DOWN, releases tag C / 1C

Digital outputs

DO	Function	Meaning / use
0	LEFT driver FWD	Direction forward
1	LEFT driver REV	Direction reverse
2	LEFT driver BRAKE	Brake
3	LIDAR AREA SEL 1	Lidar field set (set_lidar_area), OFF by default
4	RIGHT driver FWD	Direction forward
5	RIGHT driver REV	Direction reverse
6	RIGHT driver BRAKE	Brake
7	LIDAR AREA SEL 2	Lidar field set (set_lidar_area), OFF by default
8	HORN AUTO	ON while RUNNING (also auto-tune / rotate); OFF while the ALARM horn is on
9	HORN ALARM	ON the whole time the AGV is **REVERSING** (incl. the hitch hold), and when the trolley is not caught 10 s after the CC / marker stop
10	spare	—
11	AT HOMEPOST	ON while parked at homepost
12	PUSHER DOWN a	Relay K1 · with DO15
13	PUSHER UP a	Relay K2 · with DO14
14	PUSHER UP b	Relay K3 · with DO13
15	PUSHER DOWN b	Relay K4 · with DO12

Analog outputs (AO module 192.168.1.30, register base 4096)

AO	Function	Range
0	LEFT motor speed	Code sends 0–5 V (driver 0–10 V → 0–3000 rpm)
1	RIGHT motor speed	Code sends 0–5 V (driver 0–10 V → 0–3000 rpm)

Network, CAN and USB

Device	Link	Address
IPC (Ubuntu)	enp7s0 → IO modules	192.168.1.1
IPC (Ubuntu)	enp4s0 → RFID reader	192.168.3.100
DIO module	Modbus TCP	192.168.1.31:502 · unit 1
AO module	Modbus TCP	192.168.1.30:502 · unit 1
RFID reader	TCP (built-in network module)	192.168.3.200:2022
MGS1600 front	CAN 500 kbit/s (CANable2 USB)	Node 6 · TPDO 0x186
MGS1600 rear	CAN 500 kbit/s (CANable2 USB)	Node 7 · TPDO 0x187 · REVERSE only
Remote / keypad	2.4 GHz USB dongle on the IPC	“XING WEI 2.4G” (profile keypad.NAME_CONTAINS)
Web HMI	HTTP	<ipc>:5000 (also opens full-screen on the IPC screen)

6. Reverse and trolley hitch (pick-up)

Step	Trigger	What happens
1	Tag 0001 (RUNNING) — branch 2: tag 0002	Stop, wait 2 s, switch to REVERSE (skipped while ignore_rfid_1 / ignore_rfid_2 is set). Branch 2 uses only 0002 (starts right, AE / BB switch the track) and ignores 0001 / 0003; other branches ignore 0002. Tag 0003 (all branches except 2) forces the LEFT track.
2	—	Reverse at 0.15 m/s on the rear sensor. Track fixed by branch (`reverse_track_by_branch`): branch 1 (DI4) and 3 (DI6) = **right**; AE / BB are ignored for them. Branch 2, 4 / no branch: starts right, 00AE → left, 00BB → right.
2b	Tag 00BA	Arms the DI10 stop marker. Before BA the marker is ignored (logged); disarmed at every reverse start.
3	Tag 00CA	Slow to 0.1 m/s for the approach.
4	DI10 marker (only after 00BA) or tag 00CC (first wins)	Brake and hold, still in REVERSE. Set ignore_rfid_1 / 2 / 3.
5	DI9 OFF (trolley caught by the limit switch)	Wait — no timeout. **Not caught after 10 s → horn switches from AUTO (DO8) to ALARM (DO9)** until DI9 goes OFF or RESET.
6	+2 s	Pusher full UP automatically (DO13 + DO14).
7	DI9 OFF AND DI14 ON (pusher top)	Leave REVERSE, drive forward at **EXTRA_SLOW 0.15 m/s** (until the next speed tag). No timeout (RESET to abort).
8	—	Tag 00CC ignored for 5 s after moving off; tags 0001 / 0002 / 0003 ignored until tag 00CE.

7. Trolley release — tag C, or tag 1C on branch 2 (drop-off)

Branch 1, 3, 4 or no branch (e.g. a CALL START run) drop the trolley at **C (000C)**. Branch 2 drives past C and drops it at **1C (001C)**. The steps are the same:

Step	Trigger	What happens
1	Tag 000C, or 001C on branch 2 (RUNNING)	Stop and brake.
2	—	Pusher DOWN automatically (DO12 + DO15).
3	DI15 ON (pusher bottom)	Pusher stops. No timeout — the AGV waits until DI15 is ON (RESET to abort).

Step	Trigger	What happens
4	+2 s	Drive forward at SLOW.
5	Tag 0000	Stop at homepost, clear the branch, back to ARMED, DO11 ON.

8. Pusher

- UP = DO13 + DO14, DOWN = DO12 + DO15, through a 4-channel relay board wired as an H-bridge. The code never energises both pairs together.
- **Pusher button DI0** (ARMED or MANUAL only): pin at bottom → full UP; at top → full DOWN; in between → opposite of its last move; press while moving → stop.
- **Keys Z / X** (ARMED or MANUAL only): full UP / full DOWN.
- Web FULL UP / FULL DOWN stop at DI14 / DI15 or after 15 s. JOG UP / DOWN ignore both proximity sensors. PUSHER E-STOP latches until cleared.
- At startup all pusher outputs are forced OFF; the pin is not moved. Trolley limit switches are in use: “hooked” = DI9 OFF and DI14 ON.

9. Lidar safety fields (AUTO forward only, NO outputs)

Input	Object in field	Effect
DI11 outer	ON	Ease down to 60 % of the current speed
DI12 middle	ON	Ease down to 30 % of the current speed
DI13 inner	ON	Brake immediately; when clear, wait 2 s, then continue

- **OFF by default** (`lidar.ENABLED: false`). Turn it on with PARAMS → LIDAR SAFETY (engineer login, AGV stopped); that switch resets to the profile value at restart. While off, a red **LIDAR OFF** badge shows on the HOME page.
- Percentages are `OUTER_PERCENT: 60` / `MIDDLE_PERCENT: 30` in the profile.
- Outputs are normally open (`NORMALLY_CLOSED: false`): OFF = clear, ON = object. With NO wiring a broken cable reads as clear.
- Never raises speed above the tag speed. Not active while REVERSING or in MANUAL. DO3 / DO7 select the lidar field set via `set_lidar_area` (none assigned; both OFF).

10. Keyboard and USB remote control

The same keys work on the **laptop keyboard** (web MANUAL page, after clicking **KEYBOARD ON**) and on the **2.4 GHz remote / keypad** whose USB dongle is plugged into the AGV PC (found automatically, hot-plug).

Key	Action	Works in
W A S D · arrows · remote ring	Hold to drive: forward / spin left / reverse / spin right; two keys = curve; release = stop	MANUAL
1 2 3 4 5	AUTO speed APPROACH / SLOW / MEDIUM / HIGH / MAX	ARMED, RUNNING
Z / X	Pusher full UP / full DOWN	ARMED, MANUAL
R	AUTO REVERSE start (ARMED) / stop (REVERSING)	ARMED, REVERSING
Q	Switch MANUAL ↔ AUTO (overrides the DI2 selector until it is moved)	Any
7 8 9 0	Start AUTO to branch 1 / 2 / 3 / 4 (other modes: select only)	ARMED
T / Y	Rotate 180° on the tape LEFT (CCW) / RIGHT (CW); T or Y again = stop	MANUAL, ARMED
K / L	Select track LEFT / RIGHT by hand — forward track (ARMED, RUNNING) or reverse track (REVERSING); the next RFID track tag can change it again	ARMED, RUNNING, REVERSING
E	EMERGENCY STOP — latched until RESET	Any

Key	Action	Works in
ENTER · remote OK	RESET (same as DI1)	Any

- Driving uses a 400 ms watchdog: the AGV stops if the key is released, the browser loses focus, the dongle is unplugged or the link drops.
- Only the remote named in keypad.NAME_CONTAINS (“XING WEI”) is used; its keys go only to the AGV (GRAB: true), not to the AGV PC desktop. python3 debugging/list_keypads.py lists connected keyboards. Turn the remote's air-mouse off.
- The service user needs group input (build_release.sh does this). Library: evdev.

11. Manual mode, homepost and horn

- Web jog pad, keyboard or remote: forward/reverse at 0.4 m/s, turns and spins at 0.2 m/s, smooth ramps at 0.8 m/s². E-stop still cuts power immediately.
- DO11 AT HOMEPOST: ON while parked at homepost. At power-up the AGV is assumed to be at homepost.
- **Brake when stopped** (brake.ACTIVE_WHEN_STOPPED: true): DO2 / DO6 brake ON whenever the AGV stands still, in every mode incl. EMERGENCY. Set false to release the motors (push by hand).
- DO8 horn AUTO: ON while RUNNING. DO9 horn ALARM: ON while REVERSING (horn.ALARM_WHEN_REVERSING: true), and when the trolley is not caught within 10 s at the CC / marker stop (RESET silences that alarm).

Rotate 180° on the tape (rotate.py, profile section rotate)

- Start: key **T** = left (CCW), **Y** = right (CW), or the **ROTATE 180°** buttons on the MANUAL page. Allowed in MANUAL or ARMED, with tape under the front sensor and **no trolley hooked** (BLOCK_WITH_TROLLEY: true).
- The AGV spins in place at SPEED_MPS: 0.10 (wheel speed). The front sensor (7.6 cm ahead of the axle) loses the tape at ~45°, finds it again at ~135°, then the AGV creeps at CREEP_MPS: 0.04 and brakes when the tape is within CENTER_TOL_MM: 5 of centre (≈ 180°).
- Afterwards it **stops and waits**, braked, in the mode it came from. Stop a turn with T / Y again, web STOP, RESET or E-stop. Aborts (brake) if the tape is not found before ~MAX_ANGLE_DEG: 260° or TIMEOUT_S: 40 s, or if the front sensor stops responding. Lidar INNER (when enabled) pauses the turn.
- Use it on a plain straight tape, away from forks, crossings and RFID tags. Spin / creep speed, centre tolerance and the angle window are on PARAMS → ROTATE 180°.

12. Web dashboard (http://<ipc>:5000)

Page	Functions	Access
HOME	Mode, sensors, PID values, RFID, branch selector, sequence status, LIDAR OFF badge	Open
MANUAL	Jog pad, KEYBOARD panel (key list, E-stop banner, remote status), bypass buttons, pusher panel, AUTO REVERSE, ROTATE 180°	Open
IO	Live DI/DO lamps	Open
IO TEST	Force any DO ON/OFF (auto-exits when the AGV drives)	Engineer
PARAMS	Feature switches; live tuning (APPLY saves as default); **PID AUTO-TUNE**; **RUN LOGS** (CSV + plot downloads)	Engineer
Calibration	Wheel-speed calibration ramp	Engineer
ERRORS	Error log, controller restart button	Open

- **Kiosk:** after kiosk/install_kiosk.sh the AGV PC logs in automatically and opens the dashboard full-screen at power-on (Alt+F4 to leave).
- Engineer login: ENGINEER button in the bottom bar, 30-minute session. Set the password with sudo AGV_ID=agv2_kim /opt/agv2/main.bin --set-engineer-password (release) or AGV_ID=agv2_kim python3 main.py --set-engineer-password (source).
- Saved tuning: /opt/agv2/data/tuning_agv2_kim.json (release) or data/ in the source folder; it overrides the profile for those values.

12b. PID logging and auto-tune

- **Run logs:** every AUTO run and auto-tune writes a CSV (time, speed mode, raw sensor left/right, error, P/I/D, output, speed reduction, wheel rpm, gains in use) and a 3-panel plot to `analysis_plot/`. Download them from PARAMS → RUN LOGS.
- **PID auto-tune** (PARAMS, ARMED, straight tape): accelerates to the chosen mode's speed, runs a short relay test (gentle weave, up to 10 s), brakes, fits a steering model (gain, drive lag, sensor look-ahead `kinematics.SENSOR_LOOKAHEAD_M = 0.076`) and proposes KP / TD / N as Calm / Recommended / Responsive. APPLY writes them like live tuning. Every speed mode (SLOW, MEDIUM, HIGH, MAX) has its own gain set; choose **SLOW BRANCH** to tune the AA / AB / AC branch gains (runs at SLOW speed — do it on the branch tape if possible).
- Aborts on tape loss, CAN timeout, lidar inner field, RESET, emergency or STOP. RFID tags are ignored during the test.

13. RFID reader

CP-203910 / R121MU-910 UHF reader, CF frame protocol, **Active mode** (sends tags by itself). Tags: EL-UHF-TML7007M on-metal labels (Impinj Monza R6-P).

Setting	Value	Why
Work mode / interface	Active / RS232	RS232 is the reader's internal link to its built-in network module — keep it
RF power	33 dBm	Maximum
Q value	1	Fast for one tag at a time
Query interval	1 (×10 ms)	Short pause between scans
Session / filter time	S0 / 0	—
Buzzer	off	No beep on every read

- Code: frames are decoded byte-wise with length and CRC check (old “split on CF” bug lost tags containing C/F). The reader's info reply is no longer read as a tag.
- Tools (stop the controller first): `debugging/rfid_settings.py read|set|restore`, `rfid_rate_test.py`, `rfid_sweep_test.py`.
- **Installation matters most:** keep the reader and antenna away from the battery (15–20 cm or more) and from motor/battery cables; on-metal tags need metal behind them.

14. Open items

Note
• Replace all placeholder RFID tag values (incl. new 001C and 00A2) with the real scanned hex.
• Tags are missed above ~0.2 m/s on this AGV: move the reader/antenna away from the battery, or use 2–3 copies of each tag in a row.
• TITIK 4 (00AD) has no track action yet. The branch is no longer cleared at BB: fork tags passed again on the way home act again for that branch.
• Reverse track is untested: if the AGV takes the wrong track backwards, swap left/right in reverse_track_by_branch (branches 1–3) or in reverse_fork_ae / reverse_merge_bb (branch 4).
• Lidar is NO: a broken lidar cable reads as clear. NC outputs are safer if the lidar supports them.
• Pusher relay protection not added yet (UP/DOWN interlock on IO TEST, dead-time on direction change). Fit a fuse on the relay supply.
• MAX 1.0 m/s is above the motor's ~0.94 m/s capability and leaves no steering headroom. Spare channels: DO10 only.